23 February 2026

Rapid and Facile Spectrophotometric Method for the Selective Determination of Peroxymonosulfate, Peroxydisulfate and Hydrogen Peroxide in Ternary Mixtures

Anna M Dorfmüller[1], Mark M Pradja[1], Christian Modrzynski[1], Jonathan Z Bloh[1]

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Abstract

Peroxides such as peroxydisulfate (PDS), peroxymonosulfate (PMS) and hydrogen peroxide (H_2O_2) are important bulk oxidants used in wastewater treatment, bleaching or leaching. In many of these processes and during electrosynthesis, they often coexist in complex mixtures. Selective quantification of each peroxide constitutes a fundamental step towards understanding and optimizing these processes. However, achieving selectivity poses a major analytical challenge since most quantification methods rely on the oxidative properties towards indicators and interference through coexisting peroxides is common. In this study, the different iodide oxidation kinetics of PDS, PMS, and H_2O_2 were utilized to differentiate between them. Time-resolved spectrophotometry at 350 nm under optimized assay conditions was established as a prerequisite for achieving sufficient selectivity and reliable quantification. Employing low iodide concentrations and short reaction times, enable selective PMS quantification. By adding molybdate catalyst, the determination of H_2O_2 is realized. Finally, the PDS concentration is derived from the total peroxide content obtained under optimized assay conditions by subtracting the H_2O_2 and PMS concentration. The assay sensitivities reach 7.62×10^3 M^{-1} (PMS), 7.68×10^3 M^{-1} (H_2O_2) and 4.1×10^3 M^{-1} (PDS) while the limit of quantification is as low as 6.6 µM (PMS), 3.9 µM (H_2O_2) and 17 µM (PDS). The degree of interference caused by coexisting peroxides was thorougly investigated and the proposed quantification procedure was successfully applied to real electrolysis samples. The simplicity and reliability of the developed method enables high-throughput screening of electrodes, electrolytes or reaction conditions and allows quality control of processes.

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Keywords

Persulfate, Peroxide Mixtures, Selective Peroxide Analysis, Iodometry, Colorimetric Method, UV/VIS Spectroscopy, Advanced Oxidation Processes

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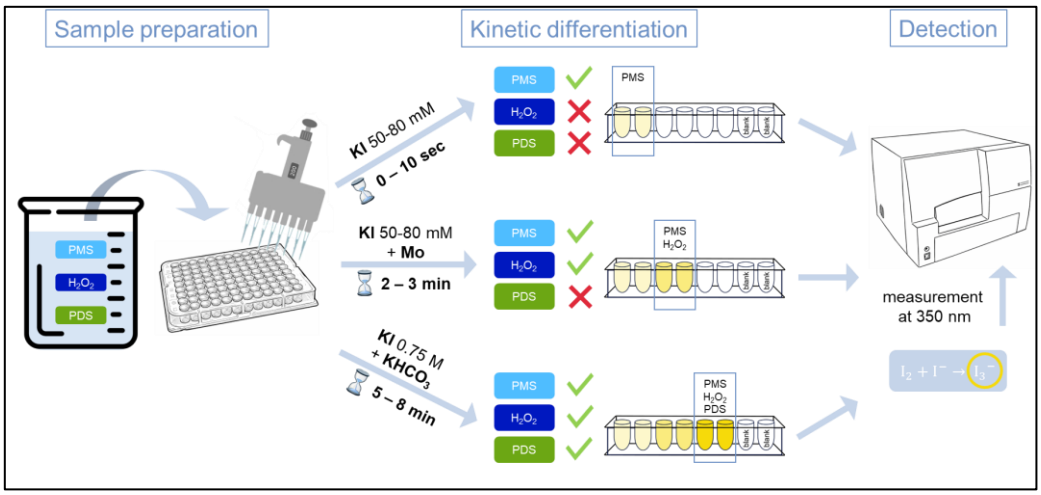
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ABSTRACT: Peroxides such as peroxydisulfate (PDS), peroxymonosulfate (PMS) and hydrogen peroxide (H_2O_2) are important bulk oxidants used in wastewater treatment, bleaching or leaching. In many of these processes and during electrosynthesis, they often coexist in complex mixtures. Selective quantification of each peroxide constitutes a fundamental step towards understanding and optimizing these processes. However, achieving selectivity poses a major analytical challenge since most quantification methods rely on the oxidative properties towards indicators and interference through coexisting peroxides is common. In this study, the different iodide oxidation kinetics of PDS, PMS, and H_2O_2 were utilized to differentiate between them. Time-resolved spectrophotometry at 350 nm under optimized assay conditions was established as a prerequisite for achieving sufficient selectivity and reliable quantification. Employing low iodide concentrations and short reaction times, enable selective PMS quantification. By adding molybdate catalyst, the determination of H_2O_2 is realized. Finally, the PDS concentration is derived from the total peroxide content obtained under optimized assay conditions by subtracting the H_2O_2 and PMS concentration. The assay sensitivities reach 7.62×10^3 M^{-1} (PMS), 7.68×10^3 M^{-1} (H_2O_2) and 4.1×10^3 M^{-1} (PDS) while the limit of quantification is as low as 6.6 µM (PMS), 3.9 µM (H_2O_2) and 17 µM (PDS). The degree of interference caused by coexisting peroxides was thorougly investigated and the proposed quantification procedure was successfully applied to real electrolysis samples. The simplicity and reliability of the developed method enables high-throughput screening of electrodes, electrolytes or reaction conditions and allows quality control of processes.

KEYWORDS: Persulfate, Peroxide Mixtures, Selective Peroxide Analysis, Iodometry, Colorimetric Method, UV/VIS Spectroscopy, Advanced Oxidation Processes



1 Introduction

Peroxydisulfate (PDS, $S_2O_8^{2-}$), peroxymonosulfate (PMS, HSO_5^-) and hydrogen peroxide (H_2O_2) are strong oxidants that are commonly used for bleaching[1, 2], polymerization[3, 4], etching[5, 6], leaching[7, 8] or wastewater treatment[9, 10]. Recently, especially advanced oxidation processes (AOPs) for wastewater treatment using persulfates (PDS and PMS) have gained increasing interest.[11, 12] Thereby, complex mixtures are commonly used, which can contain PDS, PMS and hydrogen peroxide simultaneously because binary mixtures (of PDS and H_2O_2 as well PMS and H_2O_2) have been demonstrated to be beneficial in sulfate-radical based AOPs.[10, 13, 14]

Another reason for the coexistence of several peroxides in complex mixtures is given through the fast hydrolysis (Eq. (1)) of PDS to form PMS in acidic conditions. PMS further hydrolyses forming hydrogen peroxide through Eq. (2), which is reversible and may lead to the *in situ* formation of PMS in sulfate containing peroxide solutions. As such, complex mixtures of these peroxides may arise even if only one reagent is used initially. Likewise, in electrosynthesis, the simultaneous generation of various peroxides is observed in acidic conditions [15–17] and interconversion through Eq. (1) and (2) commonly occurs during electrolysis. [18–20]

$$S_2O_8^{2-} + H^+ + H_2O \rightarrow HSO_5^- + H_2SO_4 \quad \textbf{(1)}$$

$$HSO_5^- + H^+ + H_2O \rightleftharpoons H_2O_2 + H_2SO_4 \quad \textbf{(2)}$$

These complex mixtures of PDS, PMS and H_2O_2 can be challenging to quantify due to mutual interferences because most quantification methods rely on the oxidative nature of these species towards indicators such as iodide or permanganate. Meanwhile, their chemical similarity such as the shared O-O peroxide bond, makes it challenging to distinguish between them with many analysis methods such as infrared spectroscopy. Therefore, most quantification methods target only the total peroxide concentration or study singular solutions of one peroxide and potential interference in mixtures with other peroxides is not considered. But selective quantification is important for understanding reaction mechanisms and kinetics as well as for optimizing process conditions. Commonly used detection methods include titration, UV/VIS spectroscopy, polarography or ion chromatography.[21, 22]

Concerning titration, several titrants have been reported to facilitate selective quantification of H_2O_2 in presence of PDS such as ceric sulfate[23, 24] or potassium permanganate[24]. Nevertheless, application to ternary mixtures can be problematic as the interference of PMS is unclear. More specific is the multi-step titration developed by Wang et al.[25] for the simultaneous determination of PDS, PMS and H_2O_2 in mixtures, which is based on iodometry. Nonetheless, titration often requires big sample volumes and is tedious to perform. This method also consumes the species in question, which may lead to the formation of additional PMS or H_2O_2 through the equilibrium (2) during titration, particularly since the method takes significant time. Furthermore, high relative errors arising from rapid titration, pH or temperature variations, as well as narrow quantification ranges and high detection limits, often make it unsuitable for low-concentrated samples, e.g. in electrosynthesis.[22]

Other detection methods such as polarography[26, 27] or ion chromatography[28, 29] have been proposed for the quantification of PDS but require expensive instrumentation and potential interference of H_2O_2 or PMS is often not studied. For example, H_2O_2 is expected to interfere with polarographic methods, making it unsuitable for analysing peroxide mixtures.[22] In addition, commonly used electrolytes for persulfate electrosynthesis, such as H_2SO_4 and ammonium salts, cause interference and therefore polarography cannot be used for electrolysis samples.[27]

Faster and more reliable quantification of ternary peroxide mixtures can be realized with UV/VIS spectroscopy. Multiple colorimetric reagents such as N,N-diethyl-p-phenylenediamine (DPD)[22], 2,2'-azino-bis(3-ethylbenzothiazoline-6-sulfonate) (ABTS)[30], rhodamine B[31], iodide[32] or titanyl salts[33] have been established for the quantification of peroxides. Some reagents are known to selectively react only with one peroxide, e. g. titanyl salts with H_2O_2. However, with the highly acidic measurement conditions used, the hydrolysis of PMS to H_2O_2 (Eq. (2)) is accelerated which can falsify the H_2O_2 concentration if the reaction time is too long. Other colorimetric reagents, such as iodide, are non-selective and the standard methods often estimate only the total peroxide concentration.

Nevertheless, we propose a new spectrophotometric method using iodide as the only indicator to selectively measure PDS, PMS and H_2O_2 in ternary mixtures. Iodide is a widely used reagent to quantify oxidative species photometrically or via titration. It is well known that all three peroxides (PDS, PMS and H_2O_2) react with iodide to stoichiometrically form yellow triiodide (Eq. (3) – (6)) with a maximum absorbance at 350 nm.[21–23]

$$HSO_5^- + 2I^- + H^+ \rightarrow SO_4^{2-} + I_2 + H_2O \quad \textbf{(3)}$$

$$H_2O_2 + 2I^- + 2\,H^+ \rightarrow I_2 + 2H_2O \quad \textbf{(4)}$$

$$S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2 \quad \textbf{(5)}$$

$$I_2 + I^- \rightleftharpoons I_3^- \quad \textbf{(6)}$$

Several spectrophotometric methods using iodide as indicator have already been described that can measure PMS, H_2O_2[34] and PDS[32, 35, 36] in solutions containing only one peroxide. Recently Chen et al.[21] discussed the selectivity of the photometric quantification with iodide for binary mixtures of PDS, PMS and H_2O_2. They found that with optimized KI concentrations, PMS could be selectively determined in presence of H_2O_2 or PDS, due to the much slower reaction of the latter compounds.

Here we present a newly developed method with optimized reaction conditions, which enables selective determination of PDS, PMS and H_2O_2 in ternary mixtures. Inspired by the selective determination of peracetic acid and H_2O_2 with iodide by Xiao et al.[37], this is achieved through kinetic differentiation, as the reaction rate of iodide oxidation differs significantly between the three peroxides. Furthermore, the difference in rate of reaction can be optimized through the addition of specific catalysts such as molybdate or bicarbonate. The degree of interference caused by the presence of the other peroxides was thoroughly investigated for the quantification of PDS, PMS and H_2O_2 with the proposed assays. The application of the quantification method was verified with real samples arising from electrolysis and the subsequent hydrolysis of the electrogenerated persulfates to H_2O_2 could be followed precisely with the proposed method.

2 Materials and Methods

Chemicals

Commercial hydrogen peroxide solution (30 wt%) purchased from Carl Roth GmbH was used as calibration standard as well as peroxymonosulfate in the form of Oxone®-salt purchased from Thermo Fisher Scientific and ammonium peroxydisulfate ($(NH_4)_2S_2O_8$) purchased from VWR Avantor. The H_2O_2 concentration was standardized with iodometric titration described in detail in the SI. The concentrations of the PDS and PMS stock solutions were determined with iodometric titration as well, following the method of Thompson[38]. For the spectrophotometric assays, potassium iodide (KI) and potassium bicarbonate ($KHCO_3$) were obtained from Alfa Aesar. Ammonium molybdate tetrahydrate ($(NH_4)_6Mo_7O_{24}{\cdot}4H_2O$) was purchased from Acros Organics. Potassium dihydrogen phosphate, anhydrous (KH_2PO_4) was obtained from VWR Avantor. Diluted sulfuric acid (H_2SO_4) from Carl Roth was used as electrolyte for the electrochemical experiments. All reagents were of analytical grade and were used without further purification. Ultrapure water (18 MΩ cm) was produced from a PURELAB ultra water purification system (ELGA LabWater).

Apparatus

All UV/VIS measurements were performed on a microplate reader from Agilent (PowerWave HT, formerly BioTek) using 96 well-plates (polystyrene). The electrochemical experiment was carried out in an H-cell setup with a 20 cm² BDD-anode (2 × 10 cm² front and back, expanded metal) from W&L coating systems GmbH and a carbon felt cathode as counter electrode. A Hg/Hg_2SO_4-electrode served as reference electrode in the 3-electrode setup. The distance between working and counter electrode was approximately 10 cm and the half-cells were separated through a cation exchange membrane from Fumatech BWT (F-930rfd). The anolyte and catholyte were both 100 mL of 5 M H_2SO_4. An Ivium-n-Stat potentiostat was used to apply a constant current of 2 A.

Experimental Procedure

Preparation of Stock Solutions

A 1.2 M KI stock solution was prepared by dissolving 19.9 g of potassium iodide in 100 mL of ultrapure water. The 0.1 M phosphate buffer stock solution with pH 4.0 was obtained by dissolving 68 g of KH_2PO_4 in 1 L of ultrapure water. 0.124 g of $(NH_4)_6Mo_7O_{24}{\cdot}4H_2O$ was dissolved in 20 mL of ultrapure water to obtain a 35 mM molybdate stock solution. The indicator stock mix for the KI-assay (mix 1, see Table S1), which is used to determine PMS, was prepared by mixing 14 mL phosphate buffer stock (0.5 M, pH 4.0) with 1 mL KI stock (1.2 M). The KI concentration in the obtained mix 1 is 80 mM. The indicator stock for the KI/Mo-assay (mix 2), which determines the PMS and H_2O_2 concentration, was prepared by mixing 13.83 mL phosphate buffer stock (0.5 M, pH 4.0) with 667 µL KI stock (1.2 M) and 500 µL molybdate stock (35 mM). Mix 2, as obtained, has a KI concentration of 53 mM and a molybdate concentration of 1.2 mM. Lastly, mix 3 for the KI/$KHCO_3$-assay was prepared by mixing 12.5 mL KI stock (1.2 M) and 7.5 mL $KHCO_3$ (2 M). The obtained mix 3 contains KI and $KHCO_3$ at concentrations of 0.75 M.

Under dark conditions, the stock solutions remain stable and can be stored in the refrigerator for several weeks. Note that the autooxidation of iodide cannot be completely inhibited, therefore the solutions should be freshly prepared on a regular basis. In this work the indicator stocks were discarded and reprepared when the absorbance of the blank at 350 nm was above 0.15.

PMS Determination (KI-assay)

100 µL of a sample containing various peroxides were placed in a well in a 96 well-plate. To reduce uncertainties, replicate determination was done by preparing at least two wells with the same sample. In addition, the background must also be measured *in situ* to enable baseline correction. For this purpose, 100 µL of the matrix (water or the respective electrolyte) were added to another well. Subsequently, 150 µL of mix 1 were added to all prepared wells and the absorbance at 350 nm (25 °C) was measured immediately afterwards. Note that it is important to measure the absorbance as fast as possible after the addition of indicator to selectively measure PMS concentrations. After baseline correcting the measured absorbance, the PMS concentration can be calculated using the calibration curve in Fig. 2 a). Note that calibration is specific to the employed sample container due to the employed path length and should be verified for each employed setup.

H_2O_2 Determination (KI/Mo-assay)

To determine the H_2O_2 concentration in mixtures containing PMS as well, the above-mentioned KI-assay must first be performed to measure the PMS concentration. Subsequently, the KI/Mo-assay measuring the combined concentration of PMS and H_2O_2 is proposed as follows. 100 µL of the peroxide sample were added to a well in the 96 well-plate. As with the PMS measurement described above, replicate determination should be done, and the background must be measured *in situ*. 150 µL of mix 2 were added to all the prepared wells. The absorbance at 350 nm was measured directly after the addition of indicator mix 2 for 3 minutes in 10 s intervals at 25 °C. The mean absorbance between 2 and 3 minutes was baseline corrected and used to calculate the total concentration of PMS and H_2O_2 via the calibration curve of the KI/Mo-assay (see Fig. 2 b). Based on Eq. (7), the H_2O_2 concentration can be derived from the difference between the combined concentration of PMS and H_2O_2 measured with the KI/Mo-assay and the PMS concentration measured with the KI-assay.

$$[H_2O_2] = [PMS + H_2O_2]_{KI/Mo} - [PMS]_{KI} \quad (7)$$

PDS Determination (KI/$KHCO_3$-assay)

50 µL of sample were mixed with 200 µL of mix 3 in a 96 well-plate. Replicate determination is recommended and the respective background should be measured *in situ* by mixing 50 µL of water or respective electrolyte with 200 µL of mix 3. The absorbance at 350 nm (25 °C) was measured for 8 minutes in 10 s intervals immediately after the addition of indicator. After baseline correcting the data, the mean absorbance between 5 and 8 minutes was used to calculate the total peroxide concentration using the calibration curve in Fig. 2 c). Lastly, the PDS concentration can be determined via Eq. (8) by subtracting the combined concentration of PMS and H_2O_2 measured with the KI/Mo-assay from the total peroxide concentration measured with the KI/$KHCO_3$-assay.

$$[PDS] = [PMS + H_2O_2 + PDS]_{KI/KHCO_3} - [PMS + H_2O_2]_{KI/Mo} \quad (8)$$

3 Results and Discussion

Selective Determination through Kinetic Measurements

The reaction rate of iodide oxidation differs between PDS, PMS and H_2O_2, which can be used to selectively quantify each peroxide separately. It is known that the kinetics of iodide oxidation by PMS are significantly faster than the oxidation by PDS or H_2O_2.[18, 39] Specifically, the rate of reaction between H_2O_2 and iodide (9.5×10^{-3} M^{-1} s^{-1})[34] is reported to be 6 orders of magnitude slower than the reaction rate of PMS and iodide (1.01×10^{3} M^{-1} s^{-1})[40]. In the past, the fast oxidation behaviour of PMS has been used to distinguish between PMS and PDS[38] as well as between PMS and H_2O_2[39] via iodometric titration but ternary mixtures were not investigated.

Herein, the newly developed spectroscopic method enables spectrophotometric differentiation between all three peroxides. The differentiation is achieved by selecting three different reaction conditions and catalysts which enable selective determination of either PMS alone, PMS and H_2O_2, or the total peroxide concentration. These three measurement procedures (KI-assay, KI/Mo-assay and KI/$KHCO_3$-assay) were developed for ternary mixtures of PMS, PDS and H_2O_2 and are described in detail in the following.

In Fig. 1 the absorbance at 350 nm is depicted for the three proposed assays resulting from the oxidation of iodide using equimolar amounts of PMS, PDS or H_2O_2. The absorbance is recorded with time resolution for the reaction period after the addition of indicator mix to the samples.

As can be seen in Fig. 1 a), even with relatively low KI concentrations (80 mM), PMS shows quantitative conversion already after short reaction times (0 – 10 s). This enables a selective determination of PMS in presence of both, H_2O_2 and PDS, due to their much slower reaction rates. Fig. 1 a) demonstrates that after 10 s of reaction time, equimolar amounts of H_2O_2 or PDS result in insignificantly small absorbances at 350 nm which is negligible for the PMS determination. Further studies of the interference caused by higher concentrations of H_2O_2 or PDS in mixtures with PMS are discussed in the chapter on Degree of Interference.

For a selective quantification of H_2O_2 in presence of PDS, kinetic differentiation of iodide-oxidation must be found. By adding molybdate (1.2 mM) as a catalyst to the KI-assay, the H_2O_2 oxidation is significantly accelerated, as previously observed by Copper et al.[34], while the iodide oxidation by PDS is not influenced (see Fig. 1 b). It appears that the molybdenum peroxy complex selectively forms form reacting with H_2O_2 (and possibly PMS) only, but not PDS. The reason may lie in the fact that PDS does not feature a terminal peroxide, which could form a peroxide anion and thus has a much lower nucleophilicity. This enables the determination of the combined concentration of H_2O_2 and PMS after a reaction time of 120 – 180 seconds. The chosen reaction time results from two different effects. On the one hand the reaction time needs to be sufficient to detect all H_2O_2 present, hence the plateau of absorbance must be reached, which is the case after approximately 150 s. On the other hand, the reaction time should be as short as possible to reduce inaccuracy caused by PDS slowly oxidizing iodide (see Fig. 1 b).

Lastly, a method was developed to measure the total peroxide concentration independent of the mixture composition. Thereby, the iodide oxidation rate by PDS must be increased to allow measurements in less than 15 minutes as too long reaction times lead to the decomposition of triiodide. We found that bicarbonate is a very effective, non-selective catalyst for promoting iodide oxidation, making it suitable for this purpose. The reaction rate of iodide oxidation by PDS is also strongly dependent on the KI concentration[35], thus a fast determination of the total peroxide amount can be achieved when using KI concentrations as high as 0.75 M (see the chapter on Condition Optimization). In Fig. 1 c) equal concentrations of PDS, PMS and H_2O_2 in singular solutions result in the same maximum absorbance after 300 – 480 seconds. The addition of $KHCO_3$ catalyst is essential to achieve a reliable total peroxide determination and will be discussed in the following chapter on Condition Optimization.

The absorbance spectra obtained for PMS measured with the KI-assay, for H_2O_2 measured with the KI/Mo-assay and for PDS measured with the KI/$KHCO_3$-assay are depicted in the supporting information (Fig. S1).

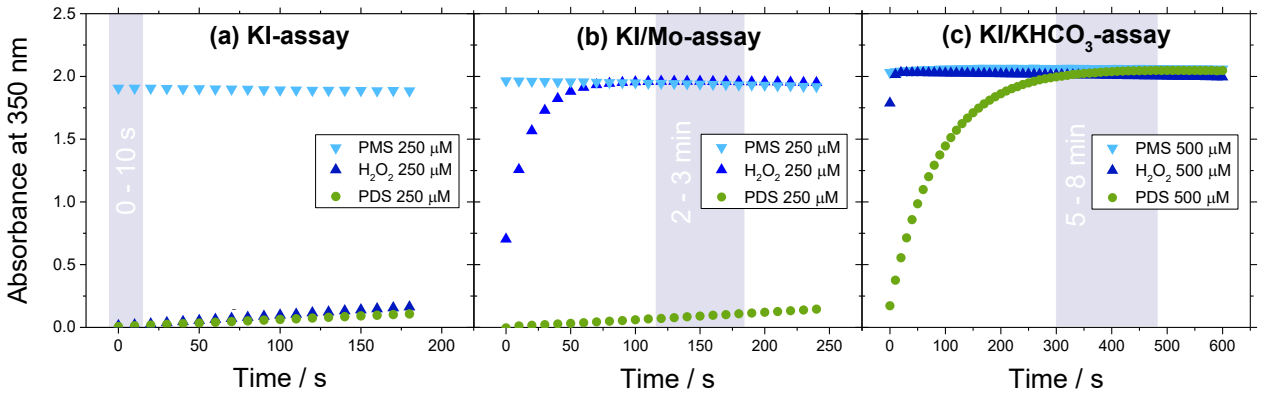


Fig. 1: Kinetic UV/VIS measurements of the absorbance at 350 nm a) using iodide at low concentrations (80 mM) to selectively measure PMS after 0 – 10 s, b) using low iodide concentrations (53 mM) and molybdate (VI) catalyst to selectively measure the combined concentration of PMS and H_2O_2 after 120 – 180 s and c) using high KI concentrations (0.75 M) and $KHCO_3$ (0.75 M) to measure the total peroxide concentration of PMS, H_2O_2 and PDS after 300 – 480 s. (Each absorbance curve results from singular solutions containing either PMS, H_2O_2 or PDS)

Calibration

Based on a PMS standard solution, calibration was performed using the KI-assay (see Fig. 2 a). The baseline-corrected absorbance at 350 nm resulting from PDS/H_2O_2 samples is also shown in Fig. 2 a) to demonstrate the selectivity of the assay. A excellent linear correlation ($R^2 = 0.9999$) between the baseline-corrected absorbance and the PMS concentration is detected for concentrations of up to 250 µM. The limit of detection (LOD) as well as the limit of quantification (LOQ) is calculated with Eq. (9) and Eq. (10) following IUPAC recommendations and ISO 11843–1.[41, 42] For the determination of PMS with the KI-assay the LOD is 2.0 µM and the LOQ is accounted to be as low as 6.6 µM. This results in a quantification range of 7 – 250 µM for PMS with a high sensitivity ($7.62 \times 10^3\ M^{-1}$).

$$\mathrm{LOD} = \frac{3 \cdot \sigma}{k} \qquad \textbf{(9)}$$

$$\mathrm{LOQ} = \frac{10 \cdot \sigma}{k} \qquad \textbf{(10)}$$

σ is the standard deviation of the mean absorbance of ten blank samples measured with the respective assay and k is the slope of the respective calibration curve (Table S2).

The calibration for the KI/Mo-assay was done using two separate standard solutions, one containing PMS and the other one containing H_2O_2 (see Fig. 2 b). Since the KI/Mo-assay allows for the simultaneous determination of PMS and H_2O_2, both peroxides in singular solutions produce the same absorbance response under the chosen assay conditions. Therefore, this procedure is effectively equivalent to replicate determinations of a single calibration curve. In the range of 4 µM up to 250 µM the joint calibration curve was linear for the detection of PMS and H_2O_2, again with an excellent coefficient of determination (R^2) of 0.9993. The quantification is highly sensitive ($7.68 \times 10^3\ M^{-1}$) with a LOD of 1.2 µM and a LOQ of 3.9 µM.

Three standard solutions containing either only PMS, H_2O_2 or PDS were used for the calibration of the KI/$KHCO_3$-assay (see Fig. 2 c). Similar to the KI/Mo-assay, this is equivalent to replicate calibration, since the KI/$KHCO_3$-assay determines the total peroxide concentration independent of the peroxide species. The equivalence of the different peroxides in the detection is again highlighted by the outstanding linear correlation ($R^2 = 0.9999$) between the absorbance at 350 nm (baseline-corrected) and the respective peroxide concentration. The concentration range in this case is 17 µM – 500 µM, the calculated LOD is 5.1 µM and the LOQ amounts to 17 µM with a high sensitivity of $4.1 \times 10^3\ M^{-1}$.

Condition Optimization

In general, the pH value is of importance for iodide assays and must be kept in an optimal range. On the one hand the pH value should not be too low because autoxidation of I^- by air (Eq. (11)) becomes a noticeable issue at low pH, potentially distorting the measurements.

$$4I^- + O_2 + 4H^+ \rightarrow 2I_2 + 2H_2O \qquad \textbf{(11)}$$

On the other hand, at pH values greater than 8, triiodide is increasingly decomposed by the disproportionation reaction shown in Eq. (12).[36]

$$3I_3^- + 6OH^- \rightarrow IO_3^- + 8I^- + 3H_2O \qquad \textbf{(12)}$$

This underlines the significance of keeping an optimal pH range which is achieved in the proposed assays (KI- and KI/Mo-assay) through the addition of phosphate buffer with pH 4.0. The KI/$KHCO_3$-assay is an exception as no phosphate buffer is used, due to kinetic reasons as well as the importance of diminishing autooxidation which will be discussed in detail in the following. Besides that, the two most important factors that need to be optimized to make selective peroxide determination possible are the reaction time and the KI concentration. For the total peroxide quantification with the KI/$KHCO_3$-assay, the $KHCO_3$ concentration also has a significant influence (see Fig. 5) and optimal conditions were investigated. The effect of the KI concentration on the PDS, PMS and H_2O_2 determination as well as the $KHCO_3$ concentration on the total peroxide determination are discussed in the following.

KI-Assay

The iodide concentration must be orders of magnitude higher than the peroxide concentration to sufficiently drive the I_2/I_3^- equilibrium Eq. (6) towards I_3^- (equilibrium constant: $768\ M^{-1}$).[43] Fig. 3 shows the baseline-corrected absorbance at 350 nm using varying KI concentrations to measure pure PMS samples as well as equimolar PMS samples in mixture with PDS

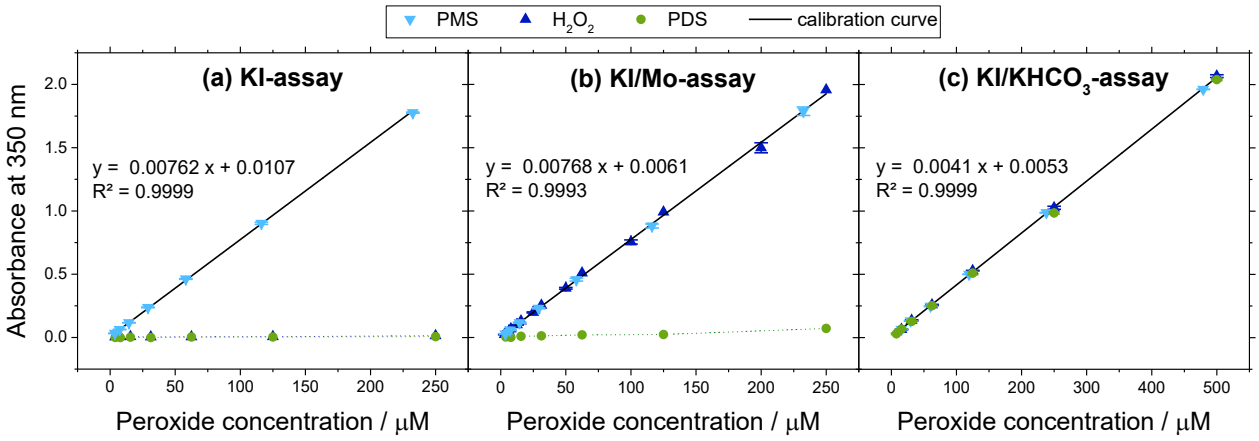


Fig. 2: Calibration curves for the selective spectrophotometric determination of (a) PMS using the KI-assay (80 mM KI) with a reaction time of 0 – 10 s, (b) PMS and H_2O_2 using the KI/Mo-assay (53 mM KI) with a reaction time of 120 – 180 s and (c) PMS, H_2O_2 and PDS using the KI/$KHCO_3$-assay (0.75 M KI) with a reaction time of 300 – 480 s. Error bars represent standard deviations (n = 2 – 3).

and H_2O_2. In Fig. 3 a) all the depicted measurements at t = 0 s resemble the initial absorbance after indicator mix 1 was added, with approximately 3 to 5 seconds of reaction time between indicator addition and first measurement. In Fig. 3 b) the same dependence of the absorbance on the KI concentration is shown for an extended reaction time of 30 s. Fig. 3 a) reveals that in absence or presence of other peroxides, the absorbance after t = 0 s increased initially with increasing KI concentration and reached a plateau for concentrations > 40 mM. Increasing the iodide concentration also accelerates the reaction rate of iodide-oxidation by other peroxides (H_2O_2, PDS), which reduces the peroxide-selectivity of the assay. This effect can be mitigated if the reaction time is kept as short as possible.

As depicted in Fig. 3 b) for an extended reaction time of 30 s, a high iodide concentration of ≥ 80 mM leads to higher absorbances when increasing amounts of other peroxides are present compared to singular PMS samples (see also Fig. S6). This leads to an overestimation of the PMS concentration, particularly if low concentrations of PMS are to be detected in a PDS/H_2O_2 excess.

Based on Fig. 3, KI concentrations between 40 and 80 mM seem to be most suitable for selective PMS determination, as there is no limitation by the I_2/I_3^- equilibrium and negligible deviations in presence of PDS/H_2O_2 are detected (reaction time of max. 10 s). In this work a KI concentration of 80 mM was chosen for the selective determination of PMS with the KI-assay after a reaction time of 0 – 10 s.

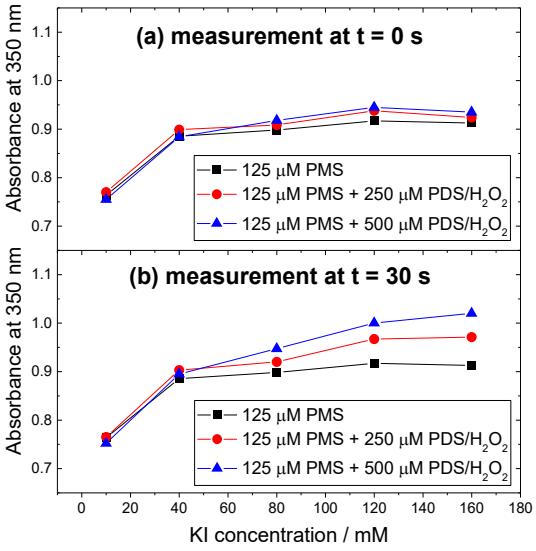


Fig. 3: Effect of different KI concentrations on the absorbance of PMS in the presence and absence of H_2O_2/PDS at (a) t = 0 s and (b) t = 30 s. The pH was kept constant at 4.0 for all measurements through the phosphate buffer.

KI/Mo-Assay

The addition of molybdate catalyst selectively accelerates the iodide oxidation by H_2O_2. This can be used to quantify the combined concentration of PMS and H_2O_2 in under 3 minutes. However, if KI concentrations below 10 mM are used, the maximum absorbance (plateau) is not reached after 2 minutes of reaction time (Fig. S2). The slower reaction kinetics at low KI concentrations lead to longer reaction times which would have higher inaccuracies due to more iodide oxidation by the slow reacting PDS. Therefore, short reaction times are desired, which is realized by using iodide concentrations > 10 mM. Similar to the KI-assay, raising the KI concentration automatically increases the kinetics of iodide-oxidation by PDS, making detection less selective when peroxide mixtures with PDS are to be analysed. Fig. 4 demonstrates that at KI concentrations ≥ 80 mM, the recorded absorbance of H_2O_2 samples in mixtures with 250 µM or 500 µM PDS is higher than for singular, equimolar H_2O_2 samples. This would lead to an overestimation of the H_2O_2 concentration. Optimal KI concentrations are therefore found to be in the range of 40 – 60 mM. In this study, 53 mM was used as KI concentration in the KI/Mo-assay for the determination of the combined concentration of PMS and H_2O_2.

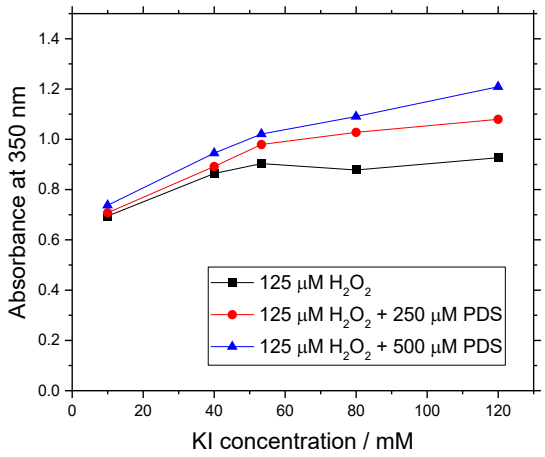


Fig. 4: Effect of the KI concentration on the mean absorbance of H_2O_2 after a reaction time of 120 – 180 s (measured in 10 s steps) in the absence and presence of different PDS concentrations at pH 4.0 with molybdate catalyst (1.2 mM).

KI/KHCO₃-Assay

To accurately determine the total peroxide concentration of complex mixtures with different composition, equimolar concentrations of PDS, PMS or H_2O_2 should yield the same absorbance after reacting with iodide for a defined time interval. As the reaction rate between iodide and PDS is relatively slow for the conditions in both assays above, higher KI concentrations were evaluated which should enhance the rate of reaction by PDS. Different KI concentrations ranging from 0.45 M to 1.05 M were tested with respect to the mean absorbance after 300 – 480 s caused by 250 µM PMS, 250 µM H_2O_2 or 250 µM PDS standard solutions (see Fig. 5 a).

With increasing KI concentration, the iodide oxidation by PDS is accelerated as can be seen in Fig. S3. When the KI concentration is below 0.75 M, the reaction rate of PDS and iodide is still too slow to reach a plateau after 8 minutes. Therefore, the mean absorbance after 300 – 480 s in Fig. 5 a) is lower for PDS than for PMS or H_2O_2. This makes accurate total peroxide determination impossible. Nevertheless, the KI concentration also shouldn't be too high to diminish autooxidation of iodide by air. Autooxidation becomes increasingly appreciable with high KI concentrations which could decrease precision. Therefore, a concentration of 0.75 M KI was selected as optimal reaction condition for the total peroxide quantification with the proposed assay.

By addition of $KHCO_3$, the autooxidation of iodide can be suppressed. This is particularly relevant when measuring samples from electrolysis, as usually strongly acidic electrolytes are used, which accelerate the autooxidation (see Eq. (11)). Fig. S4 shows the absorbance of a blank sample (50 µL H_2O) for

the time frame of the proposed assay with $KHCO_3$ (pH 8) in comparison to when phosphate buffer (pH 4) is used. It depicts the relevance of pH control to dimmish autooxidation for precise measurements.

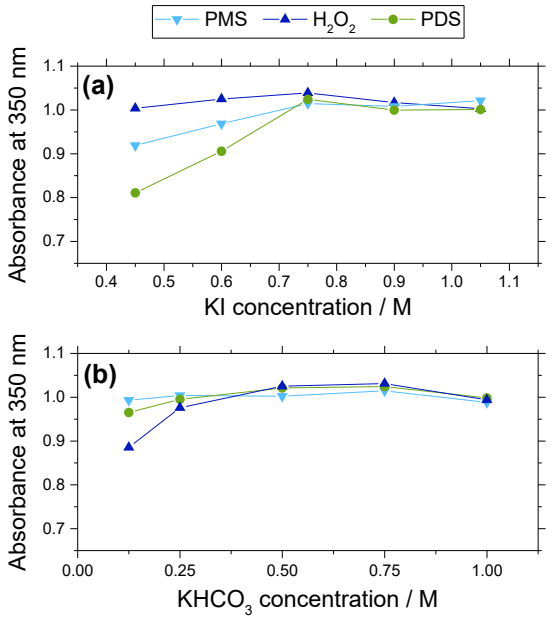


Fig. 5: Effect of (a) the KI concentration, while $KHCO_3$ was kept constant at 0.75 M, and (b) the $KHCO_3$ concentration, while KI was kept constant at 0.75 M, on the mean absorbance after 300 – 480 s (measured in 10 s steps) of 250 µM PMS, 250 µM PDS or 250 µM H_2O_2.

Furthermore, the iodide oxidation by H_2O_2 is accelerated in presence of $KHCO_3$. This is assumed to work over the intermediate peroxymonocarbonate (HCO_4^-), which is known to have faster oxidation kinetics than H_2O_2.[44] Fig. S5 depicts that in the absence of $KHCO_3$ the reaction rate of H_2O_2 is orders of magnitude slower. As depicted in Fig. 5 b), 250 µM H_2O_2 shows a significantly lower mean absorbance than equimolar concentrations of PDS and PMS when the $KHCO_3$ concentration is below 0.5 M. Thus, 0.75 M $KHCO_3$ was chosen as optimal condition.

Degree of Interference

To estimate the deviation caused by the presence of other peroxides on the selective determination with the proposed assays, the degree of interference was investigated. Using Eq. (13), the influence of other peroxides (not intended to be measured) present in mixtures on the measured absorbance is calculated.

$$\text{Degree of interference} = \left| \frac{A_{x,\,\text{mixture}} - A_{x,\,\text{single oxidant}}}{A_{x,\,\text{single oxidant}}} \right| \cdot 100\% \qquad (13)$$

$A_{x,\,\text{mixture}}$ represents the absorbance resulting from oxidant species x in mixture with other peroxides, whereas $A_{x,\,\text{single oxidant}}$ resembles the absorbance of equimolar peroxide x in the absence of other peroxides. The measurement of both $A_{x,\,\text{mixture}}$ and $A_{x,\,\text{single oxidant}}$ is repeated three times using three different samples. For the degree of interference on the PMS determination, $A_{\text{PMS,\,mixture}}$ as well as $A_{\text{PMS,\,single oxidant}}$ can be directly obtained from UV/VIS measurements with the KI-assay. The degree of interference on the H_2O_2 and PDS quantification is more complex to estimate which is described in detail in supporting information (supplementary note 2). Similar to Chen et al.[21], the degree of interference is classified into 3 regimes: no interference, minor interference and major interference. Deviations under 5% are below the limit of quantification, thus no interference was assumed for interference levels ≤ 5%. Minor interference was defined for the 5 – 10% range. A degree of interference ≥ 10% was considered to indicate major interference and was assumed unsuitable for precise peroxide quantification.

In Fig. 6 the obtained degree of interference caused by the presence of other peroxides with varying concentrations are shown for the three proposed assays. The influence of up to a 20-fold excess of other peroxides is depicted. No interference is observed for the PMS quantification (Fig. 6 a), when H_2O_2 or PDS is present in concentration up to eight times higher. Minor interference occurs only when their concentrations exceed PMS more than 10 times. Even at ratios of 1:20, the degree of interference remains minor for the PMS quantification with the proposed KI-assay. This is the same for the degree of interference by PMS on the H_2O_2 quantification with the KI/Mo-assay (Fig. 6 b).

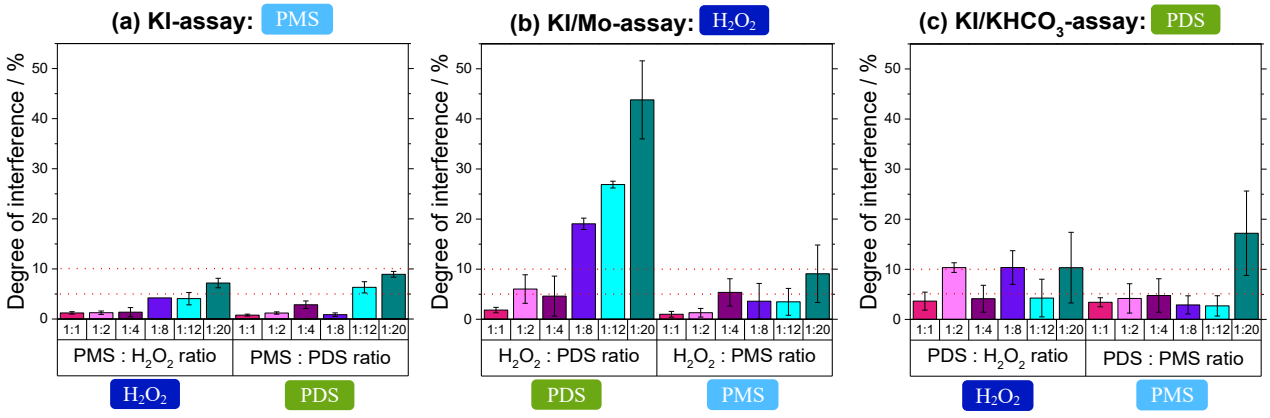


Fig. 6: (a) Degree of interference caused by H_2O_2 or PDS on the PMS detection with the proposed KI assay, (b) degree of interference caused by PDS or PMS on the H_2O_2 detection with the proposed KI/Mo-assay and (c) degree of interference caused by H_2O_2 or PMS on the PDS detection with the proposed KI/$KHCO_3$-assay. Error bars represent standard deviations (n = 3).

In contrast to that, major interference is detected for the H_2O_2 determination (Fig. 6 b) when PDS is present in 8-fold excess. At ratios of 1:20, the degree of interference is > 40% making selective H_2O_2 measurements impossible. Nevertheless, selective quantification can be achieved with the proposed KI/Mo-assay by a subsequent exponential fit of the recorded absorbance curves over the measurement time. This enables a selective H_2O_2 determination with no degree of interference even in 20-fold excess of other peroxides (see Fig. 7 b). The details of this procedure are described in the following chapter on Exponential Modelling. Alternatively, peroxide mixtures with high PDS concentrations and only traces of H_2O_2 can also be analysed selectively using titanyl sulfate ($TiOSO_4$) as indicator. A yellow peroxy complex with a maximum absorbance at 407 nm is only formed through the reaction with H_2O_2 and neither with PMS or PDS.[21, 24, 33] (see supplementary note 3) There is no degree of interference (< 5%) detected even in up to 20-fold excess of PDS or PMS (see Fig S7). The detailed quantification procedure as well as the quantification range are described in the supporting information.

Finally, Fig. 6 c) reveals that a PMS excess of ratios up to 1:12 does not interfere the detection of PDS with the KI/$KHCO_3$-assay. But a degree of interference of 17% is recorded for a PMS excess of 20 times, limiting the selective PDS quantification to PDS:PMS ratios below 1:20. H_2O_2 interferes only minorly with the PDS quantification using the KI/$KHCO_3$-assay, even at ratios of 1:20. Note that when analysing real samples with the proposed assays, replicate determination can reduce the degree of interference as over- and underestimation due to measurement variability may cancel each other out.

Exponential Modelling

The high degree of interference of the KI/Mo-assay in the presence of PDS was caused by the overlapping kinetics of its reaction with iodide which become relevant when a large excess of PDS is present. However, by fitting the recorded absorbance curves with an exponential plot combined with a linear term, the degree of interference by PDS can be minimized for the H_2O_2 quantification with the KI/Mo-assay. The curves are fitted with the following Eq. (14):

$$A(t) = A_0 \cdot \left(1 - e^{-k \cdot (t + t_0)}\right) + m \cdot t \qquad \textbf{(14)}$$

where $A(t)$ resembles the absorbance curve with A_0 being only the absorbance correlated to the iodide oxidation by H_2O_2 with the rate constant k and a time correction term t_0 that considers the time between indicator mix addition and the first absorbance measurement (usually ~ 5 s). The separate linear term with the slope m simulates the slow iodide oxidation by PDS.

Fig. 7 a) shows the fitted absorbance curves of a 125 µM H_2O_2 sample with no PDS as well as in mixture with 500 µM and 1 mM PDS. It depicts that the measured absorbance data is well described by Eq. (14) as well as the relevance of this modelling step as the mean absorbance after 120 – 180 s is significantly higher in presence of 1 mM PDS. This fit allows to separate the contributions of H_2O_2 and PDS to the total absorbances and thus enables selective H_2O_2 quantification even in highly concentrated PDS samples. This is demonstrated in Fig. 7 b), where the degree of interference is always below 5% for up to a 20-fold excess of PDS. In this case the A_0 parameter from the fit is used in place of the measured absorbance to exclude the PDS contribution.

Besides a selective H_2O_2 quantification, the exponential modelling also enables PDS determination with the KI/Mo-assay. This would be less work intensive and a faster detection method because H_2O_2 and PDS can be detected simultaneously by one assay. PDS determination is possible by using the slope m of the linear term which is only correlated to iodide being oxidized by PDS. This slope shows a linear correlation (R^2 = 0.9979) with the PDS concentration as depicted in Fig S8. The concentration of H_2O_2 does not affect the slope m obtained from the fitting (Fig. S9). But there are some limitations as this method is only applicable if the PDS concentration is larger or in the same order of magnitude as H_2O_2. When PDS is orders lower than H_2O_2, the quantification becomes inaccurate which can be seen in Fig. S10 by the large error bars of the calibration curve where samples with H_2O_2 excess where used. Traces of PDS cannot be reliably quantified using this modelling, so in this case it is better to perform the third assay (KI/$KHCO_3$) as well, to measure the PDS concentration separately.

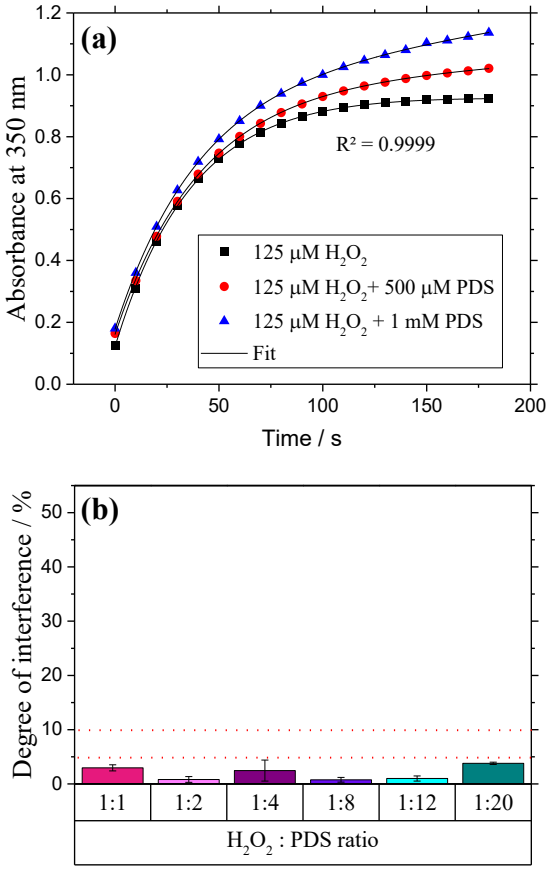


Fig. 7: (a) Fit of the absorbance curves with an exponential and linear term (Eq. (14)) for H_2O_2 samples (125 µM) with different concentrations of PDS (500 µM or 1 mM). (b) Degree of interference by PDS on the H_2O_2 quantification (KI/Mo-assay), when using the absorbance A_0 obtained from the exponential fit. Error bars represent standard deviations (n = 3).

Application to Real Samples

Understanding the peroxide synthesis through electrolysis and investigating the kinetics of the PDS, PMS and H_2O_2 interconversions during electro-generation is of importance and can be realized with the proposed detection procedure. Furthermore, the hydrolysis of PDS to PMS and subsequently H_2O_2 (Eq. (1) & (2)) following electro-synthesis has recently gained interest to enable the production of optimal peroxide mixtures for instance for leaching cathode material from lithium-ion batteries.[7]

To test the applicability of the proposed assays, 1 h electrolysis of 5 M H_2SO_4 was performed at 2 A (Fig. S11) in an H-cell setup with a 20 cm^2 (2 × 10 cm^2 front and back, expanded metal) BDD anode. Subsequently, the peroxide mixture was heated to 65 °C to promote the conversion reactions and the concentration changes through hydrolysis were detected. The PDS, PMS and H_2O_2 concentrations were measured using the KI-, KI/Mo- and the KI/$KHCO_3$-assay, respectively, by triplicate determination. For validation purposes, the $TiOSO_4$-assay was carried out as well (Fig. S12). After sampling, the samples were immediately cooled in an ice bath to stop hydrolysis and then diluted with water to adjust the concentrations to the calibration range.

After 1 h of electrolysis the total peroxide concentration reached 229 ± 4.6 mM with a current efficiency of 61% (for the calculation, see supplementary note 4). The product composition consisted of 159 ± 5.4 mM PDS, 69 ± 0.8 mM PMS and 0.77 ± 0.02 mM H_2O_2 (see Fig. 8). Afterwards the peroxide mixture was heated to 65 °C for 1.5 h and samples were taken to measure every 10 minutes for the first hour. A last sample was taken after 90 minutes.

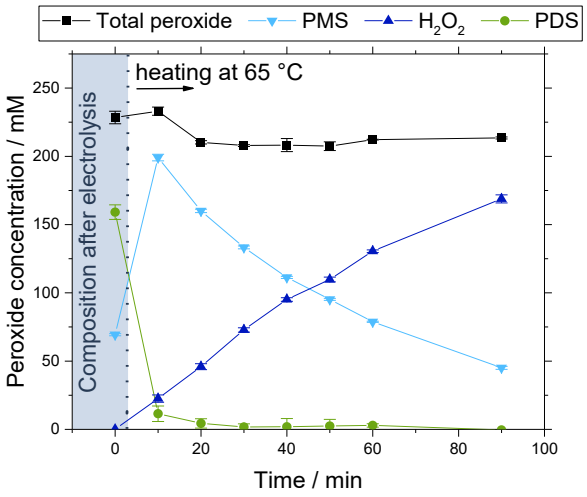


Fig. 8: Hydrolysis of an electrogenerated peroxide mixture (H-cell, 2 A, 1 h) at 65 °C, with PMS being quantified with the KI-assay, H_2O_2 with the KI/Mo-assay and PDS with the KI/$KHCO_3$-assay. Error bars represent standard deviations (n = 3).

20 minutes of heating already led to a complete conversion of PDS to PMS (Eq. (1)) and no PDS could be detected in the mixture anymore, which can be seen in Fig. 8. The fast hydrolysis to PMS was followed by the slower equilibrium reaction (Eq. (2)) to H_2O_2. After 50 minutes at 65 °C the H_2O_2 concentration exceeded the PMS concentration. When the heating was stopped the equilibrium was still not reached. The total peroxide concentration was measured to be 214 ± 0.7 mM revealing only a slight loss of peroxide due to thermal degradation. The final product composition consisted of 45 ± 1.1 mM PMS, 169 ± 2.3 mM H_2O_2 and no PDS.

This example demonstrates that the proposed assays are applicable to real samples, e.g. from electrolysis. The precise determination of the separate peroxide species also enables new insights into their formation mechanisms and kinetic studies of their interconversion rates. This method of quantification also allows quality control of processes as a simple and fast detection method of different peroxides in mixtures, without interferences in the described concentration ratios (see the chapter on Degree of Interference).

4 Conclusion

A simple, reliable and fast quantification method was established, achieving selective and precise determination of PDS, PMS and H_2O_2 in ternary mixtures. It was demonstrated that KI, as the only indicator, allows to spectrophotometrically quantify all three coexisting peroxides by kinetic differentiation with different catalysts at 350 nm. The measurement conditions have been optimized to improve the selectivity of each assay. PMS can be detected with a high sensitivity of 7.62 × 10^3 M^{-1} and a limit of quantification of 6.6 µM using the proposed KI-assay. The highly sensitive quantification (7.68 × 10^3 M^{-1}) of H_2O_2 with the KI/Mo-assay is possible for concentrations higher 3.9 µM. Based on the KI/$KHCO_3$-assay, PDS can be measured with a LOQ of 17 µM and a high sensitivity (4.1 × 10^3 M^{-1}). The degree of interference caused by the coexistence of the other peroxides in up to 20-fold excess is minor or non-detectable for the KI- and the KI/$KHCO_3$-assay as well as for the KI/Mo-assay with exponential modelling. A successful application of the proposed assays to samples arising from electrolysis was demonstrated and the results were validated with the $TiOSO_4$-assay.

The short reaction times of 0 – 10 seconds for the KI-assay, 2 – 3 minutes for the KI/Mo-assay and 5 – 8 minutes for KI/$KHCO_3$-assay enable rapid quantification. Additionally, the use of microplate photometers allows the simultaneous quantification of multiple samples, making the proposed method very time and labour efficient and enabling high-throughput screening. The use of inexpensive, common and non-toxic reagents paired with low required sample volumes and fast implementation make it an attractive tool for routine measurements of samples with various PDS, PMS and H_2O_2 compositions. Based on the developed quantification method the formation and conversion kinetics of these peroxides can easily be investigated for various reaction conditions and kinetic simulations of the peroxide electrosynthesis become possible.

Acknowledgements

The authors gratefully thank the German Federal Ministry of Education and Research (BMBF) for financial support within the project “PerDemand” (03XP0615C) and the project “E-CycLiBatt” (03XP0515).

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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